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Computer Systems Engineering

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Activity: Plenary Direct Data Addressing on Mode 3

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Most instructions can use the direct data-addressing mode.

In fact, direct data addressing is applied to many instructions in a typical program.

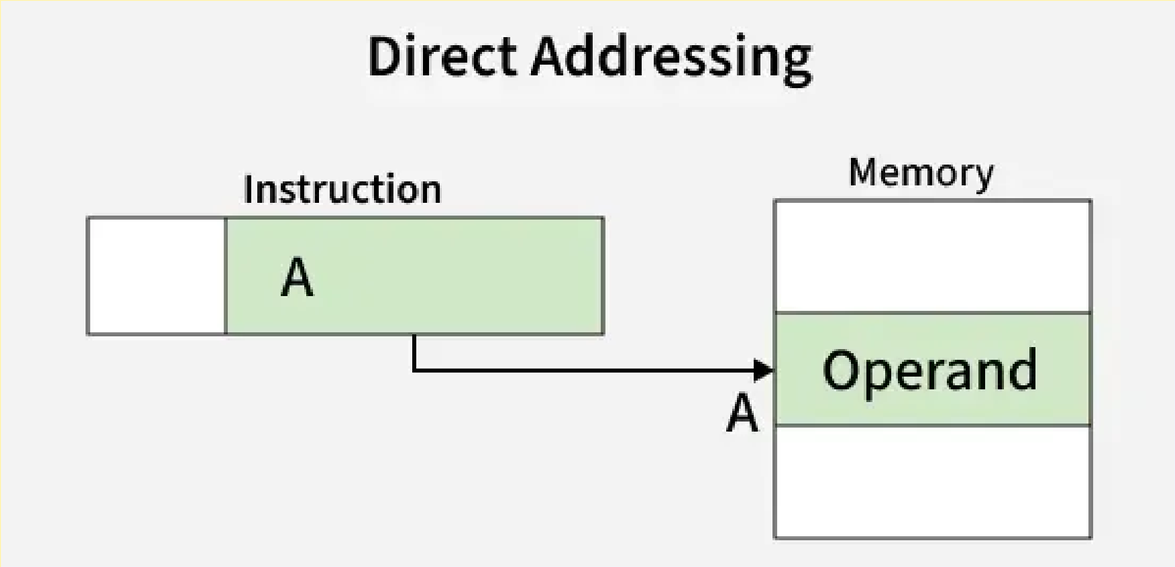
There are two basic forms of direct data addressing:

(1) Direct addressing, which applies to a MOV between a memory location and AL, AX, or EAX.

(2) Displacement addressing, which applies to almost any instruction in the instruction set. In either case, the address is formed by adding the displacement to the default data segment address or an alternate segment address.

Direct Addressing

- Applies specifically to a MOV between a memory location, located within the data segment and the AL (8-bit), AX (16-bit), or EAX (32-bit) register.
- A MOV instruction using this type of addressing is usually a special 3-byte-long instruction.
- Example: MOV AL, [1234H] which loads the byte-sized contents of memory location 1234H into register AL.
- Real Mode Address Generation: The effective address is formed by adding the offset to the data segment address times 10H. Ejemplo: DS x 10H; 10000H + 1234H = 11234H.



Displacement Addressing Explication:

Offset addressing is a routing mode in which an instruction uses a record and a displacement value to access a memory address.

It is similar to direct addressing, but offers greater flexibility because it allows you to calculate the address from a base register and a displacement

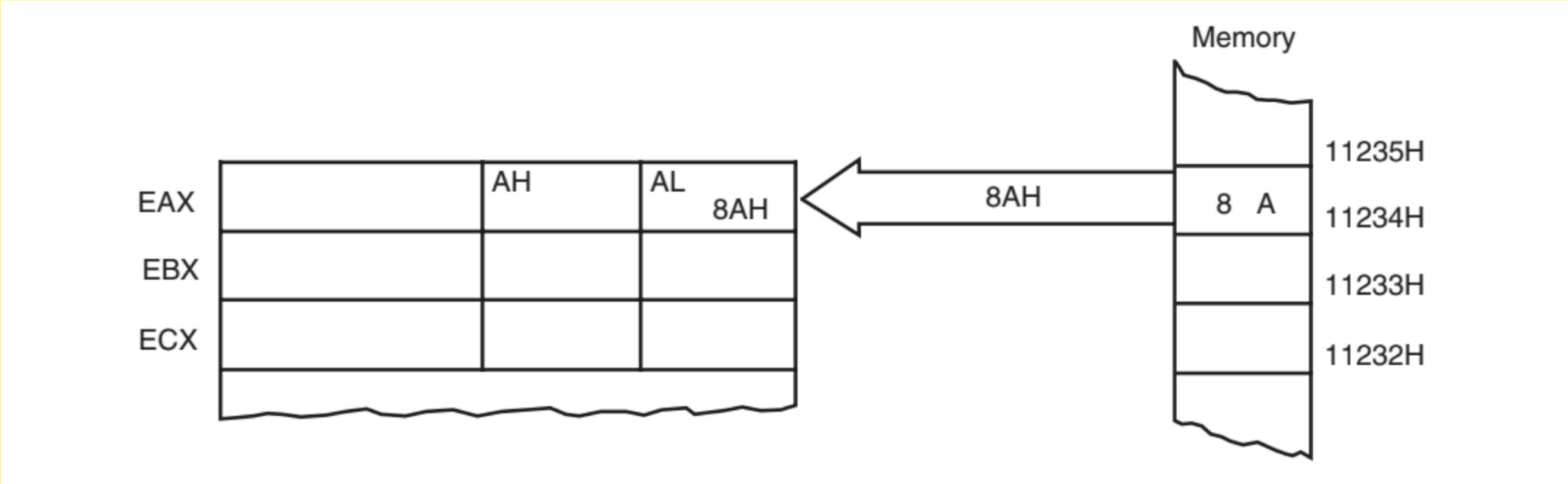
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For example, in the MOV CL instruction, DS:[1234H], a memory position indicated by the DS segment and the 1234H offset is accessed, to transfer the data to the CL register.

In the scheme we can see how the processor records relate to the memory positions and how the data that is needed is located.

Displacement Addressing:

- Almost identical to direct addressing, except the instruction is 4 bytes wide instead of 3 and can be up to 7 bytes wide in the 80386 and above if both a 32-bit register and a 32-bit displacement are specified.
- Much more flexible because most instructions use it.
- Example: MOV CL, DS:[1234H].



Direct Address Examples:

Assembly Language	*Size*	*Operation*
MOV AL,NUMBER	8 bits	Copies the byte contents of data segment memory location NUMBER into AL
MOV AX,COW	16 bits	Copies the word contents of data segment memory location COW into AX
MOV EAX,WATER*	32 bits	Copies the doubleword contents of data segment location WATER into EAX
MOV NEWS,AL	8 bits	Copies AL into byte memory location NEWS
MOV THERE,AX	16 bits	Copies AX into word memory location THERE
MOV HOME,EAX*	32 bits	Copies EAX into doubleword memory location HOME
MOV ES:[2000H],AL	8 bits	Copies AL into extra segment memory at offset address 2000H
MOV AL,MOUSE	8 bits	Copies the contents of location MOUSE into AL; in 64-bit mode MOUSE can be any address
MOV RAX,WHISKEY	64 bits	Copies 8 bytes from memory location WHISKEY into RAX

Ejemplo:

MOV ES:[2000H],AL

Assume that ES points to 3000H

3000H x 10H = 30000H

30000H + 2000H = 32000H

Origin: AL

Destino: 32000H

Displacement Addressing Examples:

Assembly Language	*Size*	*Operation*
MOV CH,DOG	8 bits	Copies the byte contents of data segment memory location DOG into CH
MOV CH,DS:[1000H]	8 bits	Copies the byte contents of data segment memory offset address 1000H into CH
MOV ES,DATA6	16 bits	Copies the word contents of data segment memory location DATA6 into ES
MOV DATA7,BP	16 bits	Copies BP into data segment memory location DATA7
MOV NUMBER,SP	16 bits	Copies SP into data segment memory location NUMBER
MOV DATA1,EAX	32 bits	Copies EAX into data segment memory location DATA1
MOV EDI,SUM1	32 bits	Copies the doubleword contents of data segment memory location SUM1 into EDI

Ejemplo:

MOV CH,DS:[1000H]

Assume that ES points to 2000H

2000H x 10H = 20000H

20000H + 1000H = 21000H

Origin: 21000H

Destino: CH

Questions

1. How is the effective memory address calculated?

2. What is an advantage of displacement addressing?